Machine Learning with Orange

Session 2: Clustering

Unsupervised Learning | Temporal Patterns

Where Were We?

✅ SUPERVISED (Session 1)

"We have labels"

Model learns: input → known output

Examples: Classification, Regression

🔍 UNSUPERVISED (Today)

"No labels — find structure"

Model learns: input → discovered groups

Examples: Clustering, Anomaly Detection

💡 **Clustering = letting the data tell you its own story**

Today's Pipeline

Data
Load dataset

→

Preprocess
Normalize, clean

→

Cluster
Find groups

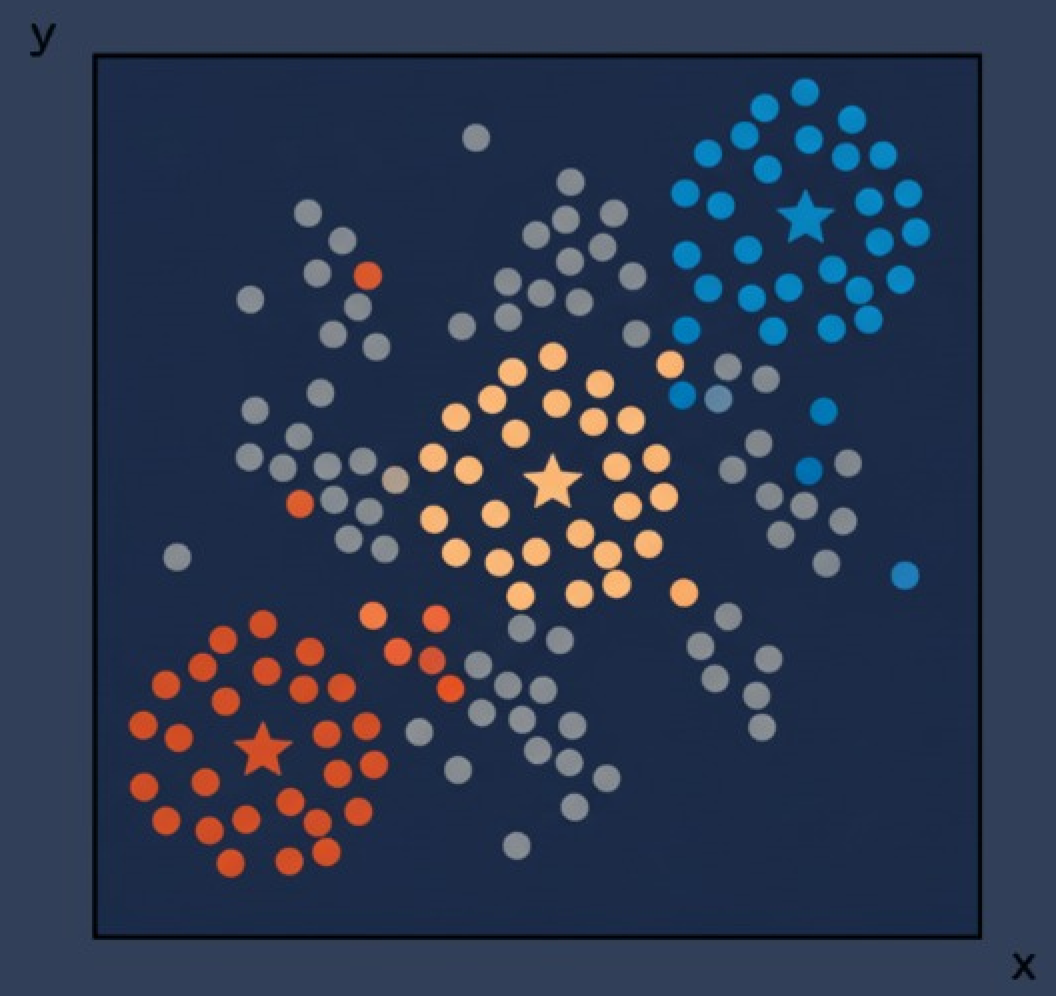
→

Interpret
Make sense

> Evaluation is different — no 'correct' labels to compare against

What is Clustering?

- Group similar data points together
- Maximize within-group similarity
- Maximize between-group difference
- ? We don't know groups in advance



Real use: customer segmentation, gene expression, document grouping

k-Means Clustering — How It Works

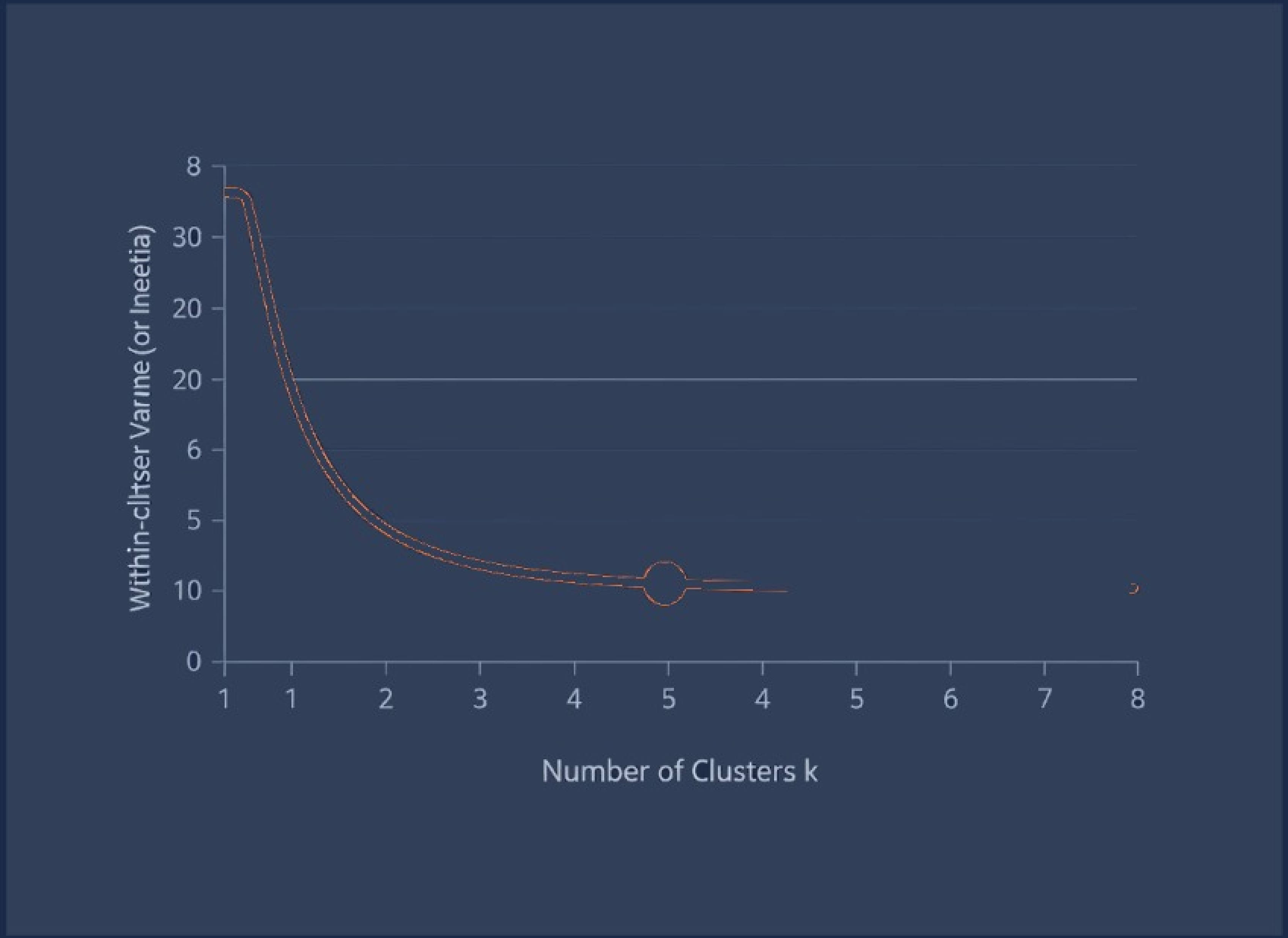
1. Pick k random centroids (★)
2. Assign each point to nearest centroid
3. Move centroid to mean of its group
4. Repeat until stable

Key Properties

- Must choose k in advance
- Works best with round, similar-sized clusters
- Sensitive to outliers

In Orange: k-Means widget — try different k values

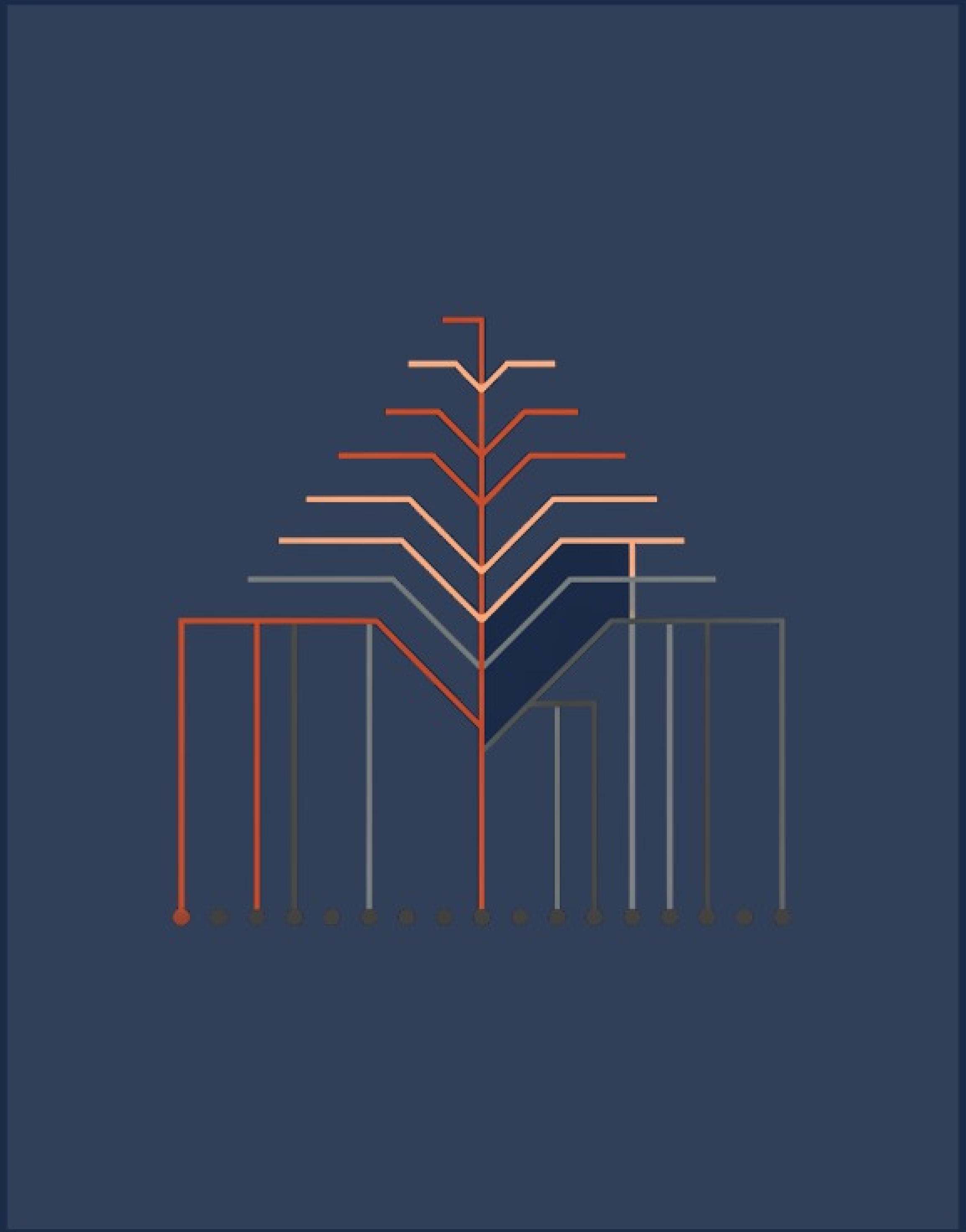
How Many Clusters? — The Elbow Method



- Too few clusters → groups are too broad
- Too many clusters → every point is its own group
- Elbow = point of diminishing returns
- Also consider: Silhouette Score for validation

💡 In Orange: Silhouette Plot widget helps you pick the right k

Hierarchical Clustering — Another Approach



k-Means

- ✅ Fast, scalable
- ✕ Must set k upfront

Hierarchical

- ✅ No k needed, shows relationships
- ✕ Slow on large datasets

In Orange: Hierarchical Clustering + Dendrogram widgets

How Do We Know Clusters Are Good?

Silhouette Score

Range: -1 to +1

High = well-separated clusters

Within-Cluster Variance

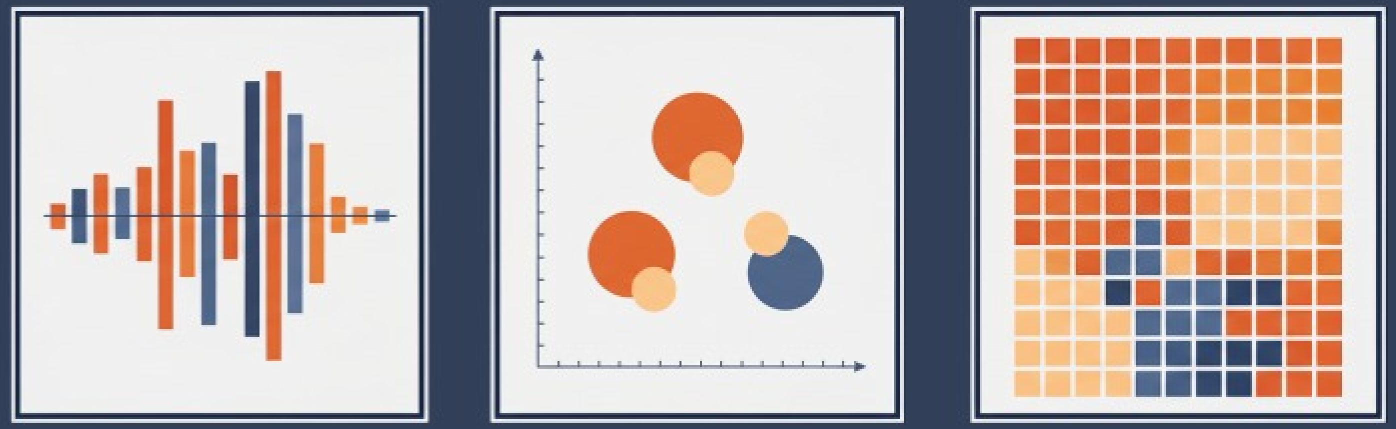
Measures cohesion

Lower = tighter, more cohesive groups

Visual Inspection

Use your eyes

Scatter plots, heatmaps — does it make sense?



No ground truth = evaluation is partly subjective. Use domain knowledge.